

Solutions and Solubility — Paper D

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must talk about **particles** — solute particles spreading between solvent particles, collisions, or how much the solvent can hold at that temperature.
- Solubility is always given in **g per 100 g of water**. If a question uses a different mass of water, scale it.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe experiments you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- Show every subtraction. A correct number with no working is one mark, not three.

SECTION 1 – AN UNFAMILIAR CURVE

1. Use the table.

Solubility (g per 100 g water)

Temperature (°C):	0	20	40	60	80	100
potassium nitrate:	13	32	64	110	169	246
copper sulfate:	14	21	29	40	55	75
sodium chloride:	36	36	37	37	38	40

- (a)** Sketch, in the space, the solubility curves for potassium nitrate and sodium chloride on one set of axes (temperature along the bottom, solubility up the side). Label both. [3]

- (b)** Estimate the solubility of potassium nitrate at 70 °C, and explain how you got it [2]
from the table.

- (c)** A saturated potassium nitrate solution in 100 g of water cools from 80 °C to 40 °C. Calculate the mass of crystals that forms. [2]

- (d)** A saturated sodium chloride solution in 100 g of water cools from 100 °C to 0 °C. Calculate the mass of crystals, and state what this tells you about recovering salt by cooling. [1]

SECTION 2 – THE LONG EXPLANATION

- 2.** A student keeps adding sugar to a glass of water at room temperature, stirring each spoonful until it dissolves. Eventually a spoonful will not dissolve, however long she stirs. She warms the glass and it dissolves. [6]

Explain everything she observed in terms of particles: what dissolving is, why the sugar stopped dissolving, why stirring could not help, and why warming did.

SECTION 3 – APPLYING THE IDEAS

- 3.** 30 g of copper sulfate is stirred into 150 g of water at 20 °C. Solubility of copper sulfate at 20 °C: 21 g per 100 g of water.

- (a) Calculate the maximum mass of copper sulfate that 150 g of water can hold at 20 °C. [2]

- (b) State whether all 30 g dissolves, and the total mass of the beaker's contents. [2]

4. Explain why dissolving is classed as a **physical** change and a solution is classed as a **mixture**. Give one piece of evidence for each. [4]

5. Nail varnish does not dissolve in water but dissolves easily in propanone. A student says "nail varnish is insoluble". Explain what is incomplete about this statement. [2]

6. Two students wrote the following in a test. Neither answer is correct.

[6]

(i) "When salt dissolves it melts into the water, so the solution weighs a bit less than the salt and water did."

(ii) "Crushing the salt lets more of it dissolve, because smaller pieces fit between the water particles."

For each one, explain the mistake and write a correct version.

End of paper. Check every answer has a unit or a form the question actually asked for.